

ARTIFICIAL INTELLIGENCE & DATA SCIENCE FOR MIS

UNIT – VI

BUSINESS AND AI

- Artificial Intelligence (AI) is becoming increasingly important in the business world as companies seek to automate processes, reduce costs, and gain insights from data. Here are some ways in which AI is being used in business:

1. **Predictive Maintenance:** Many manufacturing companies use AI to predict when their equipment will fail and proactively perform maintenance. For example, General Electric uses sensors and machine learning to predict when aircraft engines will need maintenance, saving airlines billions of dollars in unplanned downtime.
2. **Personalized Marketing:** Retail companies like Amazon and Netflix use AI algorithms to analyze customer data and recommend personalized products or content. This improves the customer experience and increases sales.
3. **Fraud Detection:** Financial institutions use AI to detect and prevent fraud, such as credit card fraud and money laundering. For example, PayPal uses machine learning to identify fraudulent transactions and prevent financial losses.

BUSINESS AND AI

4. Chatbots: Many businesses use chatbots powered by AI to provide instant customer support. For example, H&M uses a chatbot to help customers with product recommendations and sizing questions.
5. Autonomous Vehicles: Companies like Uber and Tesla are developing self-driving cars that use AI to navigate roads and avoid obstacles. This technology has the potential to revolutionize the transportation industry.
6. Sentiment Analysis: Companies use AI to analyze customer feedback on social media and online reviews. This helps them understand customer sentiment and improve their products and services. For example, Airbnb uses machine learning to analyze customer reviews and improve the quality of its listings.
7. Supply Chain Optimization: Retailers like Walmart use AI to optimize their supply chain operations, such as inventory management and logistics. This reduces costs and improves efficiency

AN OVERVIEW OF ARTIFICIAL INTELLIGENCE

- Artificial Intelligence (AI) is a type of technology that allows machines to perform tasks that normally require human intelligence, such as learning, problem-solving, decision-making, and perception.
- It involves the development of algorithms and computer programs that can analyze data, recognize patterns, and make predictions or decisions based on that data.
- AI encompasses a range of technologies, including machine learning, natural language processing, robotics, and computer vision.

MACHINE LEARNING

- Machine learning is a subset of AI that focuses on the development of algorithms that enable computers to learn from data. These algorithms can be trained to recognize patterns in large datasets and make predictions or decisions based on those patterns. Machine learning is used in a wide range of applications, from fraud detection and predictive maintenance to recommendation systems and autonomous vehicles.

NATURAL LANGUAGE PROCESSING (NLP)

- Natural language processing (NLP) is a branch of AI that focuses on the interaction between computers and human languages. NLP technology enables computers to understand, interpret, and generate human language, which is used in applications such as virtual assistants, chatbots, and language translation.

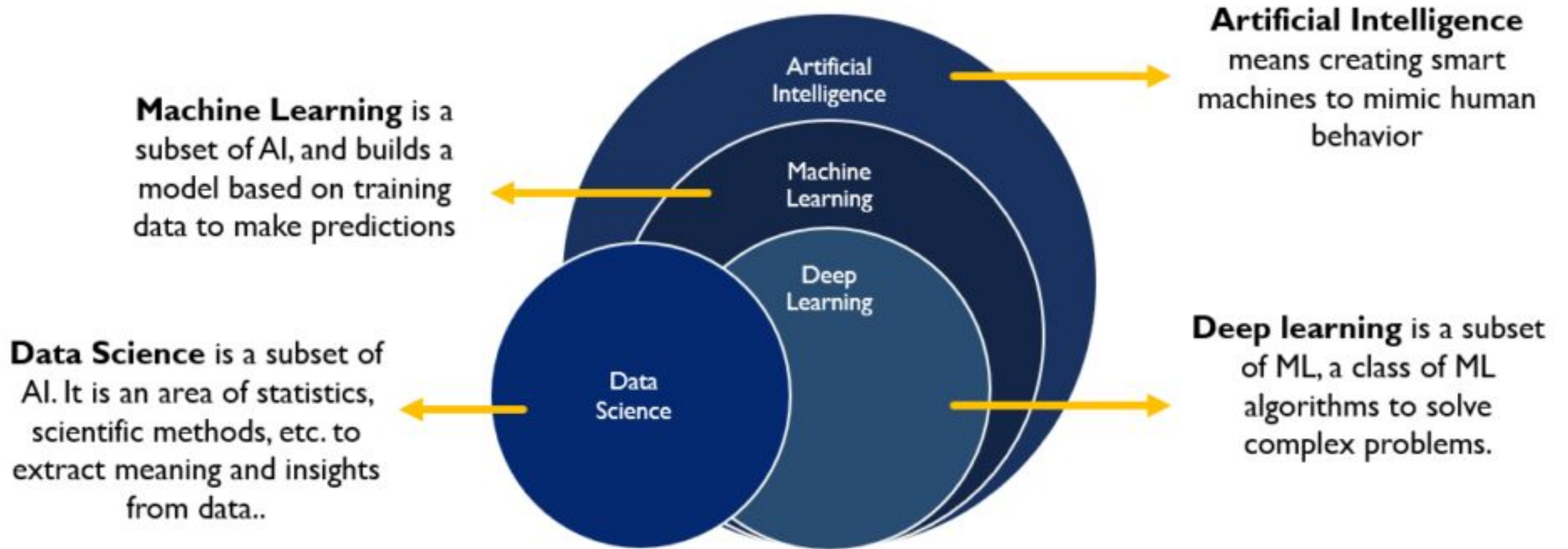
ROBOTICS

- Robotics is the development of machines that can perform tasks autonomously or semi-autonomously. Robotics technology is used in manufacturing, healthcare, and other industries to perform tasks that are dangerous, repetitive, or require a high level of precision.

COMPUTER VISION

- Computer vision is the development of algorithms that enable computers to interpret and analyze visual data, such as images and videos. Computer vision technology is used in applications such as facial recognition, object detection, and autonomous vehicles.

VENN DIAGRAM



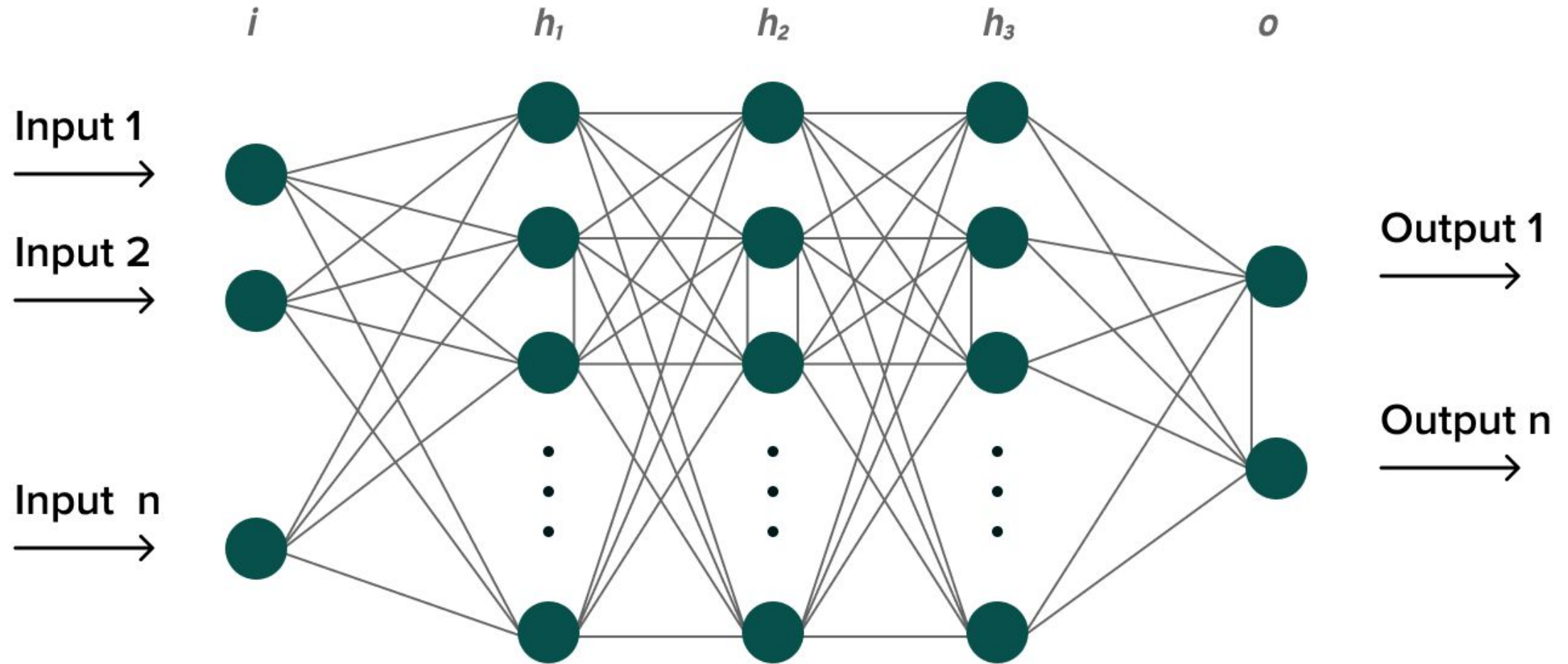
NEURAL NETWORK

- A neural network is a method in artificial intelligence that teaches computers to process data in a way that is inspired by the human brain.
- It is a type of machine learning process, called deep learning, that uses interconnected nodes or neurons in a layered structure that resembles the human brain.
- It creates an adaptive system that computers use to learn from their mistakes and improve continuously.
- Thus, artificial neural networks attempt to solve complicated problems, like summarizing documents or recognizing faces, with greater accuracy.

Input layer

Hidden layers

Output layer



SIMPLE NEURAL NETWORK ARCHITECTURE

- A basic neural network has interconnected artificial neurons in three layers:
 - Input Layer
 - Information from the outside world enters the artificial neural network from the input layer. Input nodes process the data, analyze or categorize it, and pass it on to the next layer.
 - Hidden Layer
 - Hidden layers take their input from the input layer or other hidden layers. Artificial neural networks can have a large number of hidden layers. Each hidden layer analyzes the output from the previous layer, processes it further, and passes it on to the next layer.

SIMPLE NEURAL NETWORK ARCHITECTURE

- Output Layer

- The output layer gives the final result of all the data processing by the artificial neural network. It can have single or multiple nodes. For instance, if we have a binary (yes/no) classification problem, the output layer will have one output node, which will give the result as 1 or 0. However, if we have a multi-class classification problem, the output layer might consist of more than one output node.

VIRTUAL REALITY (VR)

- Virtual reality (VR) refers to a computer-generated simulation of a three-dimensional environment that can be experienced through a headset or other devices. The simulation is designed to be immersive, so users feel like they are actually present in the virtual world.
- Virtual reality systems typically use specialized hardware, including a headset with a display screen or screens, sensors to track the user's movements, and input devices such as controllers or hand-held trackers. These devices work together to create a seamless and realistic virtual environment for the user to interact with.

VIRTUAL REALITY (VR)

- Virtual reality has a wide range of applications across different industries, including gaming, education, healthcare, and engineering.
- In gaming, VR allows players to fully immerse themselves in a game environment, creating a more engaging and interactive experience. I
- n education, VR can be used to provide students with immersive simulations of historical events, scientific concepts, or real-world scenarios.
- In healthcare, VR is used for therapy and rehabilitation, as well as training medical professionals.
- In engineering, VR can be used for virtual prototyping and design review.

FUZZY LOGIC

- Fuzzy means uncertain, indefinite, or unclear.
- Fuzzy logic is a computing technique that is based on the degree of truth.
- A fuzzy logic system uses the input's degree of truth and linguistic variables to produce a certain output.
- The state of this input determines the nature of the output.

FUZZY LOGIC

- This technique is different from boolean logic, which uses only two categories (true or false).
- In boolean logic, two categories (0 and 1) are used to describe objects.
- For example, the temperature in water served in glass may be High (1) or Low (0).
- The water is described using more categories in fuzzy logic, but within the two categories mentioned earlier. In this case, the water may be very cold, very warm, or warm.

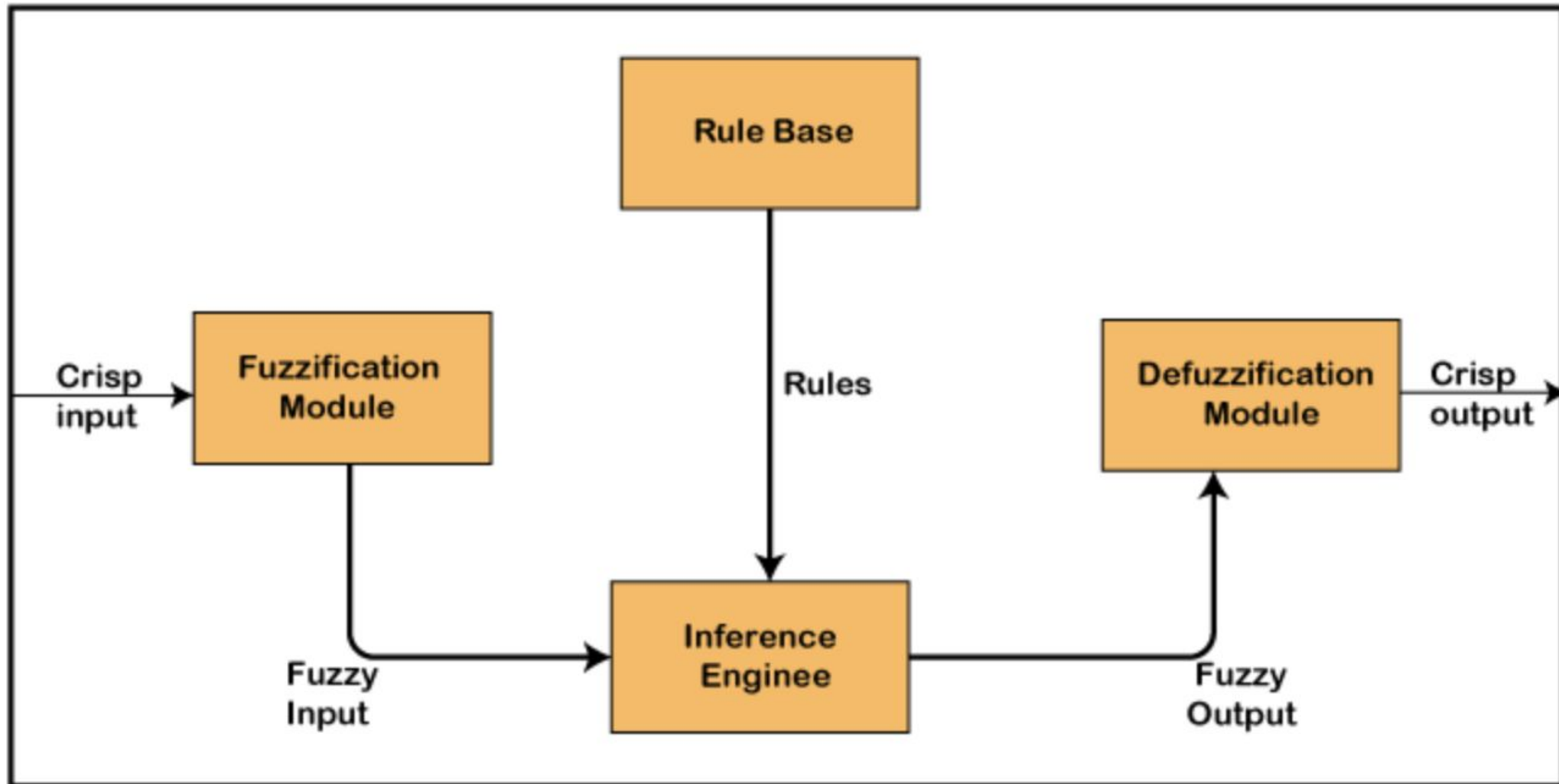
FUZZY LOGIC

- In boolean logic, the answer would be either yes or no. In fuzzy logic, the answer may be between these two categories.
- Some of the probable answers in this logic may include possibly yes, possibly no, or certainly no.

WHY FUZZY LOGIC IS USED

- It solves the problem of uncertainty in the engineering field.
- When accurate reasoning is not available, it provides an accurate level of reasoning.
- Fuzzy logic has a simple structure that is easy to understand.
- It is an effective way of controlling machines.
- It provides solutions to various industrial problems (especially decision making).
- It requires little data to be executed.

FUZZY LOGIC ARCHITECTURE

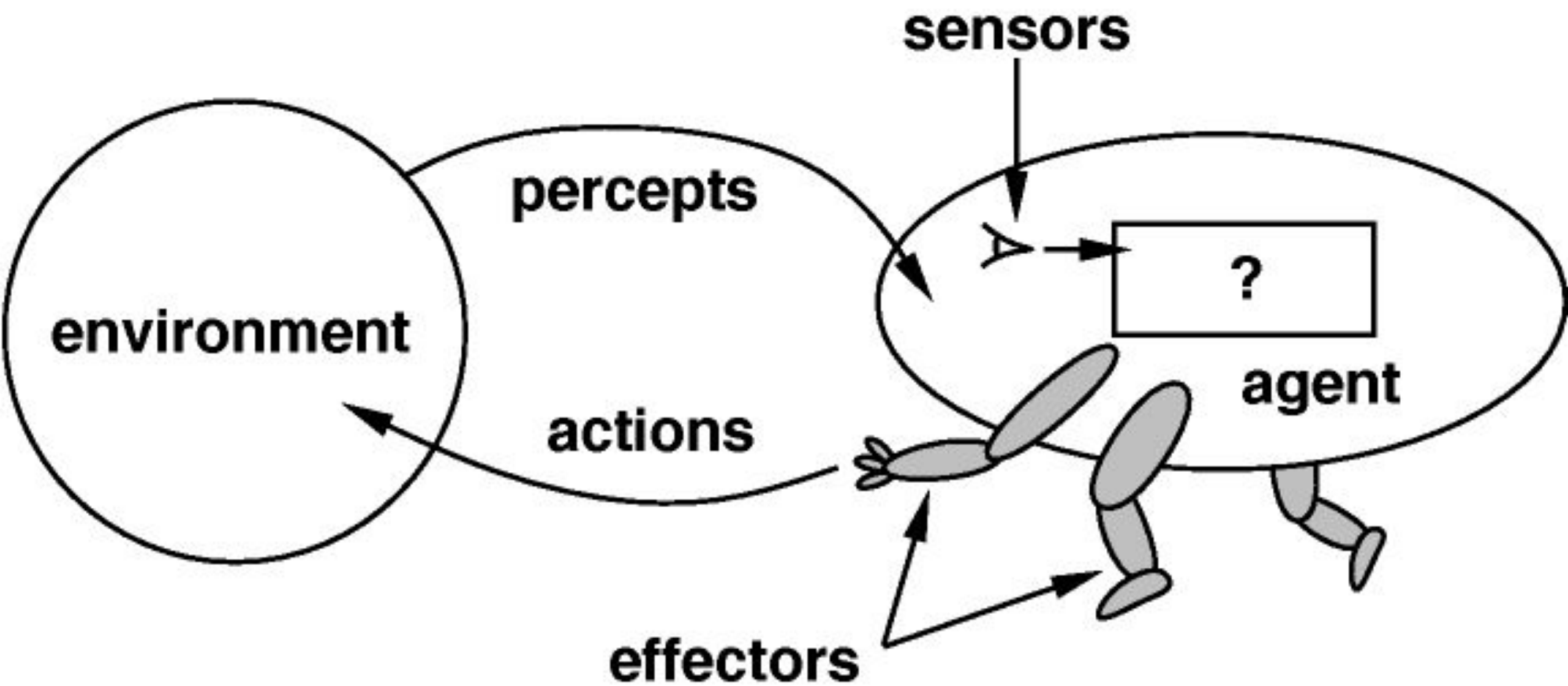


FUZZY LOGIC ARCHITECTURE

- Rule Base: This contains the rules and membership functions that regulate or control decision-making in the fuzzy logic system. It also contains the IF-THEN conditions used for conditional programming and controlling the system.
- Fuzzifier: This component transforms raw inputs into fuzzy sets. The fuzzy sets proceed to the control system, where they undergo further processing.
- Inference Engine: This is a tool that establishes the ideal rules for a specific input. It then applies these rules to the input data to generate a fuzzy output.
- Defuzzifier: This component transforms the fuzzy sets into an explicit output (in the form of crisp inputs). Defuzzification is the final stage of a fuzzy logic system.

INTELLIGENT AGENT

- An intelligent agent is a program that can make decisions or perform a service based on its environment, user input and experiences.
- These programs can be used to autonomously gather information on a regular, programmed schedule or when prompted by the user in real time.
- Intelligent agents may also be referred to as a bot, which is short for robot.
- In agents that employ artificial intelligence (AI), user input is collected using sensors, like microphone or cameras, and agent output is delivered through actuators, like speakers or screens.
- The practice of having information brought to a user by an agent is called push technology.



TYPES OF INTELLIGENT AGENTS

- Reflex agents: These agents function in a current state, ignoring past history. Responses are based on the event-condition-action rule (ECA rule) where a user initiates an event and the agent refers to a list of pre-set rules and pre-programmed outcomes.
- Model-based agents: These agents choose an action in the same way as a reflex agent, but they have a more comprehensive view of the environment. A model of the world is programmed into the internal system that incorporates the agent's history.
- Goal-based agents: These agents expand upon the information model-based agents store by also including goal information, or information about desirable situations.

TYPES OF INTELLIGENT AGENTS

- Utility-based agents: These agents are similar to goal-based agents but provide an extra utility measurement which rates each possible scenario on its desired result and chooses the action that maximizes the outcome. Rating criteria examples could be the probability of success or the resources required.
- Learning agents: These agents have the ability to gradually improve and become more knowledgeable about an environment over time through an additional learning element. The learning element will use feedback to determine how performance elements should be changed to improve gradually.

APPLICATIONS

1. **Personal Assistants:** Intelligent agents can act as personal assistants by understanding natural language queries and performing tasks such as scheduling appointments, sending messages, making reservations, providing recommendations, and managing personal information.
2. **Customer Service:** Intelligent agents are used in customer service applications to interact with customers, answer frequently asked questions, provide support, and assist with troubleshooting common issues. They can be deployed as chatbots or virtual assistants on websites, mobile apps, or messaging platforms.
3. **Autonomous Vehicles:** Intelligent agents play a crucial role in autonomous vehicles by perceiving the environment, analyzing sensor data, making decisions, and controlling the vehicle's actions. They help with tasks like lane detection, object recognition, path planning, and collision avoidance.
4. **Smart Homes:** Intelligent agents can control and manage various devices and systems within a smart home, including lighting, heating, security cameras, appliances, and entertainment systems. They can learn from user preferences and automate routine tasks to enhance comfort, convenience, and energy efficiency.
5. **Healthcare:** Intelligent agents are employed in healthcare settings to assist with tasks like diagnosis, treatment planning, medication management, monitoring patient conditions, and providing personalized health recommendations. They can leverage medical knowledge and patient data to improve healthcare outcomes.

APPLICATIONS

6. Finance and Trading: Intelligent agents are used in financial applications for tasks such as analyzing market data, predicting trends, portfolio management, risk assessment, and executing trades. They can leverage machine learning algorithms to make informed investment decisions.
7. Supply Chain Management: Intelligent agents can optimize supply chain processes by monitoring inventory levels, forecasting demand, managing logistics, and coordinating activities across multiple stakeholders. They can improve efficiency, reduce costs, and enhance overall supply chain performance.
8. Gaming: Intelligent agents are employed in computer games to provide realistic and challenging opponents, assist players with hints or guidance, and dynamically adapt game difficulty based on player performance. They can simulate human-like behavior and enhance the gaming experience.
9. Fraud Detection: Intelligent agents can be used to detect and prevent fraudulent activities in various domains, including banking, insurance, and e-commerce. They can analyze patterns, detect anomalies, and raise alerts for suspicious transactions or behaviors.
10. Virtual Assistants: Intelligent agents like Siri, Google Assistant, or Alexa act as virtual assistants on smartphones or smart speakers. They can perform tasks such as voice recognition, natural language processing, web searches, playing music, setting reminders, and controlling smart home devices.

EXPERT SYSTEMS

- Expert systems are computer-based systems that utilize knowledge and reasoning capabilities to solve complex problems in a specific domain. They are designed to mimic the decision-making process of human experts by capturing and utilizing their knowledge. Here's an example of an expert system:
- Example: Medical Diagnosis Expert System
- The medical field is one domain where expert systems have been extensively used. An expert system can assist doctors in diagnosing diseases based on patient symptoms, medical history, and test results. Let's consider an example of a medical diagnosis expert system for respiratory illnesses:

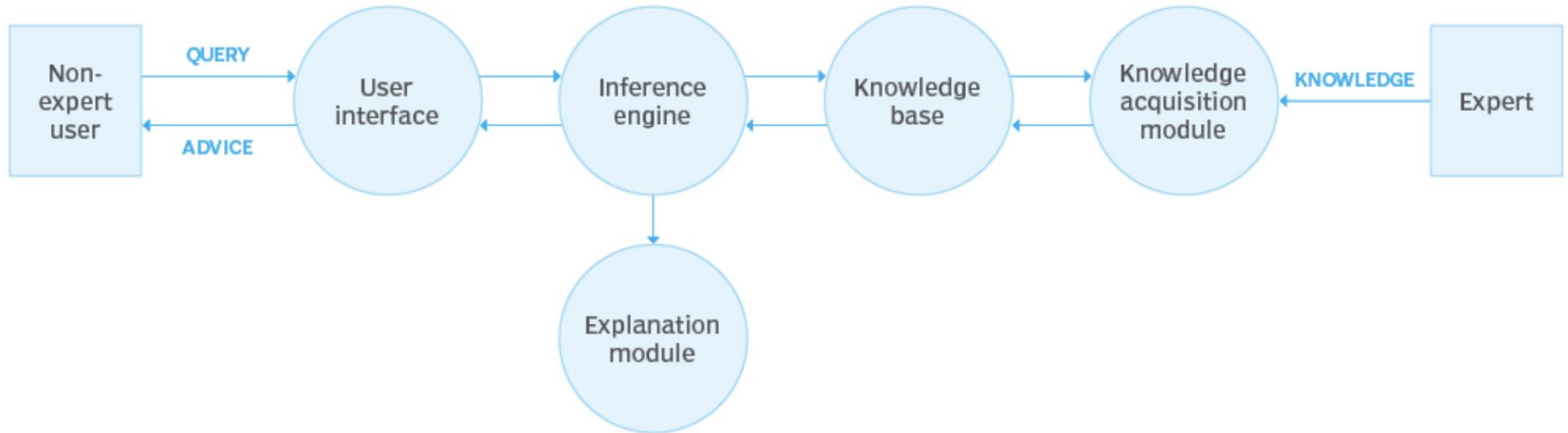
EXPERT SYSTEMS

1. Knowledge Acquisition: The expert system is developed by gathering knowledge from medical experts in the field of respiratory illnesses. This knowledge is typically acquired through interviews, surveys, or by analyzing medical textbooks and research papers.
2. Knowledge Representation: The acquired knowledge is represented in the form of rules or a knowledge base. For instance, the expert system may have rules like:
 1. IF patient has a cough AND patient has a fever AND patient has difficulty breathing THEN diagnose the patient with pneumonia.
 2. IF patient has a cough AND patient has wheezing THEN diagnose the patient with asthma.
 3. IF patient has a cough AND patient has runny nose AND patient has sore throat THEN diagnose the patient with common cold.

EXPERT SYSTEMS

4. Inference Engine: The expert system utilizes an inference engine to reason and make deductions based on the rules and the given symptoms. The inference engine uses forward chaining or backward chaining to arrive at a diagnosis. It asks the user a series of questions about the symptoms and uses the provided information to narrow down the possible diagnoses.
5. User Interface: The expert system has a user interface that allows doctors or patients to input the symptoms and receive a diagnosis. The interface can be text-based or graphical, depending on the system's design.
6. Explanation and Recommendations: After making a diagnosis, the expert system can provide an explanation of the reasoning process and the factors that led to the specific diagnosis. It can also suggest appropriate treatments, medications, and further tests if necessary.

Architecture of an expert system



EXPERT SYSTEMS

- Forward chaining reads and processes a set of facts to make a logical prediction about what will happen next. An example of forward chaining would be making predictions about the movement of the stock market.
- Backward chaining reads and processes a set of facts to reach a logical conclusion about why something happened. An example of backward chaining would be examining a set of symptoms to reach a medical diagnosis.

BENEFITS OF EXPERT SYSTEMS

- Knowledge capture and preservation.
- Consistent decision making.
- Scalability and availability.
- Enhanced training and support.
- Increased efficiency and productivity.

LIMITATIONS OF EXPERT SYSTEMS

- Lack of common sense and intuition.
- Limited adaptability to new or unforeseen situations.
- Difficulty in handling uncertain or incomplete information.
- High development and maintenance costs.
- Lack of user trust and acceptance.

GENETIC ALGORITHM

- A genetic algorithm is a search and optimization technique inspired by the process of natural selection and evolution. It is used to find optimal solutions to complex problems by iteratively evolving a population of candidate solutions. Here's an overview of how genetic algorithms work:
1. Initialization: A population of potential solutions, often represented as a set of chromosomes or strings of genes, is randomly generated. Each chromosome represents a candidate solution to the problem.
 2. Evaluation: Each chromosome in the population is evaluated and assigned a fitness value based on how well it solves the problem. The fitness function determines the quality or fitness of a solution, which can be based on various criteria or objectives.

GENETIC ALGORITHM

3. Selection: Chromosomes with higher fitness values have a higher chance of being selected for the next generation. Selection methods such as tournament selection or roulette wheel selection are used to choose individuals from the population to undergo genetic operations.
4. Genetic Operations:
 - a. Crossover: Two selected chromosomes are combined to create new offspring. Crossover involves exchanging genetic material (genes) between parents to create a new solution. Different crossover techniques, such as one-point crossover or uniform crossover, determine how genes are exchanged.
 - b. Mutation: Random changes are introduced to the genetic material of selected chromosomes. Mutation helps introduce diversity into the population by randomly altering individual genes or characteristics of a chromosome.

GENETIC ALGORITHM

5. Offspring Generation: The new offspring created through crossover and mutation replace a portion of the old population. The size of the population remains constant to maintain diversity.
6. Evaluation and Termination: The fitness of the new population is evaluated, and the process continues with selection, crossover, and mutation for multiple generations or until a termination criterion is met. The termination criterion can be a maximum number of generations, reaching a satisfactory solution, or a predefined threshold for the fitness value.
7. Solution Extraction: Once the algorithm terminates, the best solution or solutions found during the process, typically those with the highest fitness values, are extracted as the optimal or near-optimal solutions to the problem.

APPLICATIONS

- **Transportation and Logistics:** Genetic algorithms are applied in optimizing transportation and logistics operations. They help in solving problems like route optimization, vehicle scheduling, fleet management, and cargo loading to minimize costs, reduce travel time, and maximize efficiency.
- **Energy Management:** Genetic algorithms aid in optimizing energy systems by determining the optimal scheduling of power generation, load balancing, and resource allocation. They are used in renewable energy integration, smart grid management, and demand-side management.
- **Manufacturing and Production:** Genetic algorithms optimize production processes by addressing issues such as production scheduling, inventory management, resource allocation, and production line optimization. They help in improving productivity, reducing costs, and minimizing downtime.

APPLICATIONS

- Finance and Investment: Genetic algorithms assist in portfolio optimization, where they determine the optimal allocation of investments across various assets. They consider risk, return, and diversification to create investment portfolios that maximize performance and minimize risk.
- Telecommunications: Genetic algorithms optimize network routing, resource allocation, and frequency assignment in telecommunications systems. They help in improving network performance, reducing congestion, and maximizing the utilization of resources.
- Medicine and Healthcare: Genetic algorithms are used in medical applications like drug discovery, treatment optimization, and personalized medicine. They assist in identifying optimal drug combinations, treatment protocols, and genetic biomarkers associated with diseases.

APPLICATIONS

- **Data Mining and Pattern Recognition:** Genetic algorithms aid in data analysis tasks such as pattern recognition, feature selection, and clustering. They help in identifying meaningful patterns, reducing data dimensionality, and improving classification or prediction accuracy.
- **Image and Signal Processing:** Genetic algorithms are utilized in image and signal processing applications such as image compression, image reconstruction, signal filtering, and feature extraction. They optimize parameters and configurations to enhance the quality of processed images and signals.
- **Architecture and Design:** Genetic algorithms are applied in architectural design to optimize building layouts, floor plans, and structural configurations. They help in finding designs that meet specific criteria such as energy efficiency, structural stability, and aesthetic appeal.

MIS IN DATA SCIENCE: TRANSITION INTO DATA SCIENCE FOR A BUSINESS INTELLIGENCE (BI)/MIS PROFESSIONAL

- <https://www.analyticsvidhya.com/blog/2020/09/11-steps-data-science-transition-reporting-mis-bi-professionals/>
- <https://medium.com/analytics-vidhya/11-steps-to-transition-into-data-science-for-reporting-mis-bi-professionals-622da22692a1>
- Only 6 steps